# AMIRHOSSEIN GHOLINIA

## Python Developer | AI Application & Agent Developer

**Professional Summary**

Python Developer specializing in AI application development, AI Agent engineering, and integrating Large Language Models (LLMs) into software applications.

I have hands-on experience designing AI Agent architectures with LangGraph, connecting language models to custom tools through Tool Calling, and developing intelligent applications using Python.

My experience includes building Semantic Search systems and recommendation solutions using machine learning techniques, as well as combining LLM capabilities with custom software logic to create practical AI-powered applications.

I am particularly interested in AI Agents, intelligent automation, and developing software solutions powered by modern AI technologies.

**Technical Skills**

**1. AI Application Development**

- AI-Powered Application Development
- LLM Integration
- AI Model API Integration
- Tool Calling
- Connecting LLMs to Custom Tools and Functions
- AI Chatbot Development
- Intelligent Automation

**2. AI Agent Development & Engineering**

- AI Agent Development
- LangGraph
- LangChain
- Agent Execution Flow Design
- Custom Tools and Tool Calling
- Agent Workflows
- State and Conversation Management

**3. Semantic Search & Text Processing**

- Semantic Search
- Sentence Transformers
- Transformers
- Vector Embeddings
- FAISS
- Similarity Search and Retrieval

**4. Machine Learning & Recommendation Systems**

- Scikit-learn
- Collaborative Filtering
- LightFM
- Recommendation Systems
- Hybrid Recommendation Systems
- Combining Semantic Search with Machine Learning

**5. Software Development & Infrastructure**

- Python
- Object-Oriented Programming (OOP)
- FastAPI
- RESTful APIs
- Pydantic
- MySQL
- PyMySQL
- Git and Version Control
- Unit Testing and Integration Testing
- Debugging

- Modular Software Architecture
- Docker (Basic Knowledge)

**6. AI Development Tools**

- Cline
- AI Coding Agents
- AI-Assisted Software Development
- Reviewing, Debugging, and Testing AI-Generated Code

**Featured Project**

## AFilm Bot — AI-Powered Movie & TV Series Recommendation Agent

Designed and developed an intelligent Telegram bot that provides personalized movie and TV series recommendations through an AI Agent architecture.

**Key Features & Implementation**

- Designed an AI Agent architecture using LangGraph.
- Integrated LLMs with custom application tools through Tool Calling.
- Implemented agent execution flows and conversation state management.
- Developed a hybrid recommendation pipeline combining Semantic Search, Collaborative Filtering, and business rules.
- Implemented semantic retrieval using Sentence Transformers and FAISS.
- Used LightFM to generate personalized recommendations based on user interactions.
- Integrated an LLM to process recommendation results and generate final suggestions.
- Designed a MySQL database structure for managing users, conversations, messages, and interactions.
- Developed the Telegram interface using the Telegram Bot API.

**Technologies:** Python, LangGraph, LangChain, LLM APIs, FAISS, Sentence Transformers, LightFM, MySQL, PyMySQL, Telegram Bot API.

**Education**

Computer Engineering Student

University of Science and Technology of Babol

**Languages**

- Persian: Native
- English: Intermediate